



Control system for quantum computers on FPGA

Quantum Instrumentation Control Kit, or QICK, a revolutionary control and readout electronics system developed by experts at the US Department of Energy's Fermi National Accelerator Laboratory, has proven to dramatically improve quantum computer performance while lowering



Gustavo Cancelo led a team of Fermilab engineers to create a new compact electronics board. It has the capabilities of an entire rack of equipment that is compatible with many designs of superconducting qubits at a fraction of the cost (Credit: <https://news.fnal.gov>)

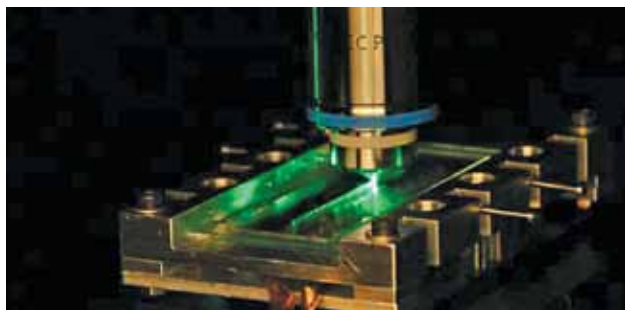
control equipment costs. A single electronics board, slightly larger than a laptop, houses the capability of a complete rack of equipment. Microwave pulses are used to control and read qubits. The Fermilab team's radio frequency (RF) board has over 200 components, including mixers, filters, amplifiers, attenuators, and switches. The RF board, when combined with a commercial FPGA board, gives scientists everything they need to connect successfully with the quantum realm.

A new wireless charging system for buses

Researchers from Taiwan's National Chiayi University have created a revolutionary UAV based charging method for city-wide sensor networks that use public transportation like buses. In this system, the bus provides the energy required for effective UAV operation, while flight energy consumption is kept to a minimum. The research contributes to the advancement of this technology in various ways. First, a method for coordinating bus schedules and UAVs has been established. Second, charging algorithms based on real-world bus data and metropolitan maps have been developed to increase UAV recharging and sensor power supply. Finally, the authors analysed three different charging tactics for the suggested system.

Sapphire fibre sensor that can withstand over 2000°C

Researchers at Oxford University have invented a sapphire fibre sensor that can withstand severe temperatures, with the potential to enhance efficiency and reduce emissions in aircrafts. The researchers have used a sapphire optical fibre—a thread of industrially grown sapphire less than

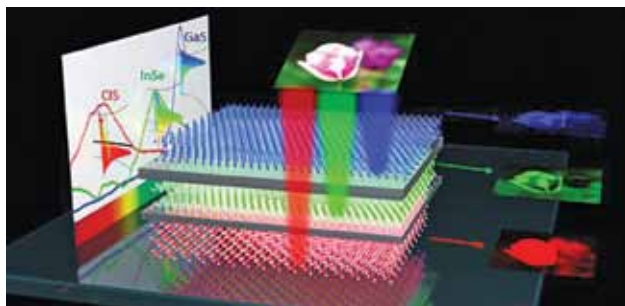


Sapphire optical fibre (Credit: <https://phys.org>)

half a millimetre thick—which can withstand temperatures of over 2000°C. The first demonstration used a 1cm-long sapphire fibre, but the researchers believe that lengths of up to several metres will be conceivable, with multiple sensors spaced along the length. This would, for example, allow temperature data to be taken throughout a jet engine.

An electric eye with a vertical colour sensor

Researchers at Georgia State University have created a new form of artificial vision device that uses a novel vertical stacking architecture to enable higher colour recognition depth and micro scalability. According to the team, the van



Development of electric eye (Credit: <https://news.gsu.edu>)

der Waals semiconductor-empowered vertical colour sensing structure has exact colour identification capabilities, which simplifies the design of an optical lens system for artificial vision. Compared with conventional semiconductors, such as silicon, this sensor can precisely control the van der Waals material band structure, thickness, and other critical parameters to sense red, green, and blue colours.